



Armada Kalamunda Group

Small for Gestational Age and Fetal Growth Restriction Guideline

1. Purpose

To inform clinicians of the screening, management and obstetric birth considerations for pregnancies complicated with Fetal Growth Restriction (FGR), also known as Intrauterine Growth Restriction (IUGR) and suspected Small for Gestation Age (SGA)

2. Applicability

This guideline applies to all medical and midwifery staff working at Armada Kalamunda Group (AKG).

3. Background

3.1 Small for gestational age fetus

The term 'small for gestational age' (SGA) refers to the fetus that has failed to reach a specific biometry or estimated weight threshold by a specific gestational age. It is estimated that **50-70%** of foetuses born weighing less than the 10th centile for gestational age are constitutionally small, with the growth appropriate for the parental size and ethnicity. The outcome is usually associated with normal placental function and normal outcomes. SGA foetuses with a birth weight less than the 50th centile for gestational age have a greater likelihood of FGR.

SGA foetuses are at greater risk for stillbirth, birth hypoxia, neonatal complications, impaired neurodevelopment, and possibly Type 2 diabetes and hypertension in adult life, although the high incidence of adverse perinatal outcomes maybe contributed to the FGR foetuses in this group. The majority of term SGA infants have no appreciable morbidity or mortality.

Biometric tests used to assess fetal size assist diagnosis of SGA, while biophysical tests are used to detect fetal wellbeing and are more indicative of FGR. The use of the customised Fundal Height chart has been demonstrated to improve the accuracy to predict a SGA foetus, but ultrasound measurements of the abdominal circumference and estimated fetal weight provide the most accurate way to predict SGA. Fundal Height (FH) measurements may improve sensitivity and specificity for predicting SGA, whilst abdominal palpation alone has limited accuracy for identification of an SGA fetus. The impact on perinatal outcomes of FH measurement, compared to abdominal palpation, is uncertain with a Cochrane systematic review finding only one controlled trial that showed FH measurements did not significantly change perinatal outcomes. Continuation of FH measurement at each antenatal appointment has been recommended.

Assessment of fetal growth, abdominal circumference (AC) and estimated fetal weight (EFW), requires two ultrasound measurements at least two to three weeks apart, which will differentiate normally growing foetuses from those with FGR.

Routine biometry is not justified in third trimester as it does not reduce the risk of SGA and does not improve perinatal outcomes. Measurements only provide limited information to assist decision making for management for timing of delivery. Associated antenatal surveillance techniques assist in clinical judgement for timing of delivery. These techniques differentiate between a SGA fetus with a predicted normal outcome, and the fetus which is growth restricted resulting in adverse perinatal morbidity and mortality. Umbilical artery (UA) Doppler measurements can identify if a confirmed SGA fetus is affected by placental dysfunction, with end-diastolic flow velocity results providing valuable information on risk for perinatal mortality and morbidity.

3.2 Key Points/ Definition of SGA:

1. SGA describes the fetus that has failed to reach the normal biometric weight by a specific gestational age. This does not always indicate a fetus is growth restricted.
2. The use of ultrasound biometry and biophysical tests can assist differentiation between the SGA with no expected perinatal morbidity or mortality risk, and the FGR fetus with predicted poor perinatal outcomes.
3. To evaluate fetal growth over time at least two subsequent measurements two weeks apart should be performed. A three week interval further reduces false positive results.
4. Management is individualised according to gestation, fetal wellbeing and any compounding maternal or fetal health factors.
5. SGA is defined as baby with Estimated Fetal weight (EFW)/ Abdominal circumference (AC) $\leq 10^{\text{th}}$ centile on customised chart with normal doppler studies

3.3 Risk factors for IUGR/SGA:

- Maternal age, >40
- Nulliparity
- BMI <20
- IVF pregnancy
- Daily vigorous exercise
- Low maternal weight gain
- Previous stillbirth
- Pre- eclampsia (previous pregnancy or this pregnancy)
- Maternal or paternal SGA
- Pregnancy interval (<6 months or >60 months)
- Heavy bleeding (threatened miscarriage), unexplained APH, or Placental abruption
- Echogenic fetal bowel
- Caffeine >300 mg/day in third trimester
- PAPP- A < 0.4 MoM
- Smoking
- Multiple pregnancy

3.4 Assess for causes of the SGA and/or the FGR fetus

- Constitutionally small mothers
- Poor maternal nutrition leading to a malnourished and underweight mother
- **Previous birth of an SGA baby** increases risk in a subsequent pregnancy

- Fetal structural abnormalities and congenital malformations
- Fetal chromosomal abnormalities
- Multiple pregnancy - a twin pregnancy is associated with a 10% increased chance of IUGR
- Life style factors e.g. smoking (**>11/day**), alcohol and substance abuse (cocaine)
- Fetal infections e.g. cytomegalovirus, malaria, parvovirus, rubella
- Maternal disease or disorders e.g. **pregnancy induced hypertension**
- (mild/ **severe**); **diabetes**; **vascular disease**; **chronic HTN**
- Disorders of cartilage and bone
- Teratogens
- **Renal disease**
- Chronic hypoxia
- Placental and cord abnormalities
- Antiphospholipid Antibody Syndrome

Note: Factors in bold represent major risk factors for IUGR

3.5 Methods to detect SGA include:

- Most methods to detect SGA require an accurate estimation of gestation as a prerequisite
- Measurement of Fundal Height –recommended at each antenatal appointment from 26 weeks improving prediction of SGA fetus.
- Abdominal palpation – has a limited diagnostic ability to predict the SGA fetus. If an SGA fetus is suspected, diagnosis should be supplemented by ultrasound biometry.
- Ultrasound biometry (AC or EFW ≤ 10 th centile).
- Doppler studies: Umbilical Artery and Middle Cerebral Artery(MCA) doppler studies

3.6 Management: (Appendix 2)

At booking identify those needing increased monitoring:

- Where FH is less accurate (large uterine fibroids, $> \text{BMI}$) = serial growth ultrasounds.
- One major or three minor risk factors present (see below).

Consider preventative interventions in high risk groups (smoking cessation advice, antiplatelet agents in women at high risk of pre-eclampsia).

4. Fetal surveillance

4.1 Ultrasound scans

1. If severe SGA identified on anatomy scan, arrange a referral to MFM team at KEMH for a detailed anatomical ultrasound and Uterine Artery Doppler
Offer karyotyping in severe SGA with structural anomalies, those before 23 weeks gestation, particularly if UA Doppler normal
2. Arrange ultrasound assessment if an SGA fetus is **suspected** – biometry, Maximum Vertical Pocket (MVP) or amniotic fluid index (AFI), Umbilical Artery (UA) Doppler and Middle Cerebral Artery (MCA) doppler studies
3. If SGA is **confirmed** organise serial assessment of fetal biometry and Doppler as follows:

- Weekly ultrasounds including AFI and UA & MCA Doppler's. Doppler is the primary surveillance tool in SGA.
 - o If **normal** Doppler flow: may be repeated every 14 days. More frequently in severe SGA
 - o If **abnormal** UA & MCA Doppler flow indices and birth not indicated, in the first place **refer to KEMH or FSH depending on gestation age**, in case of delay in appointment at Tertiary Hospital or patient refuses to Tertiary Hospital repeat Dopplers:
 - Twice weekly if end-diastolic velocities present.
 - Daily if absent/reversed end-diastolic frequencies.
- Fortnightly fetal biometry and fetal well-being.
- 4. Preterm SGA fetus with *abnormal* Doppler studies should be referred to Tertiary center as soon as possible.
- 5. In the *term* SGA fetus with *normal* UA Doppler, the middle cerebral artery (MCA) Doppler should be used to assist in timing birth or aim to deliver the baby after discussion with on call specialist

4.2 Cardiotocograph monitoring (CTG)

- If SGA is confirmed perform a CTG if the fetus is > 32 weeks gestation.
- If SGA is confirmed and the fetus is < 32 weeks gestation – discuss management with the Consultant Obstetrician if CTG monitoring is required in correlation with ultrasound findings.
- Frequency of follow-up CTG monitoring in Antenatal Assessment Unit (AAU) should be weekly for SGA babies with no risk factors and twice weekly with risk factors. The Consultant Obstetrician/ GPO must be involved in management plan especially if association with risk factors.
- The CTG should be used in conjunction with other fetal monitoring for the SGA fetus.

5. Medical review and antenatal care:

5.1 Confirmed SGA

1. Abnormalities of ultrasound examination or CTG monitoring should have timely review by the Consultant Obstetrician on call or by the Consultant Obstetrician whose Clinic the woman attends depending on the situation.
2. Document a management plan on the AKMR 70.5 (Obstetric Risk Assessment form), DMR and AAU assessment form
3. Organise post ultrasound follow-up appointments with the Consultant Obstetrician Clinic or AAU depending upon the situation. Ideally the appointments should be made to coincide with the appointments in AAU for CTG monitoring.
4. Organise CTG monitoring according to gestation and medical management plan.
5. Aim to deliver SGA pregnancy after 39 weeks if surveillance is appropriate and within normal parameters, any deviation to be discussed with on call team for timing of delivery.

When escalating care, e.g., USS or CTG concerns. The reporting midwife **MUST** co-sign with the medical officer the acknowledgement of results reviewed and/or any follow up care planned.

5.2 Confirmed Fetal Growth restriction

1. If IUGR is diagnosed refer to Section Fetal Growth restriction (FGR)/Intrauterine Growth Restriction below
2. Consider administration of corticosteroids if pre-term delivery is anticipated within a week.

6. Fetal Growth restriction (Intrauterine growth restriction)

6.1 Background

50-70% of the Small-for-Gestation Age (SGA) fetuses are constitutionally small but healthy. Approximately 10-15% of SGA fetuses are classified to be 'true' FGR/IUGR cases, and another 5-10% are associated with chromosomal/structural anomalies, or chronic intrauterine infection.

A fetus is considered to have intrauterine growth restriction when the ultrasound fetal measurements, particularly the abdominal circumference or serial weight measurements, are below what is considered normal for that age and gestation. This is usually below the 5th or 10th centile when compared to the normal growth and gestational age by ultrasound measurements. The FGR infant has not reached their genetic growth potential due to a pathological reason or events in utero causing placental dysfunction. The FGR fetus is associated with an increased risk of perinatal mortality and morbidity and long term health consequences for survivors.

Current evidence suggests long term consequences for FGR infants are that they are prone to heart disease, type 2 diabetes, strokes, hypertension and even osteoporosis later in life

Umbilical artery (UA) Doppler measurement is a tool used to identify if the SGA fetus is affected by placental dysfunction which occurs with the FGR fetus. Any SGA baby with EFW or AC ≤ 10th centile with abnormal UA Doppler studies (UAPI > 95th centile and MCAP < 5th centile) is diagnosed as IUGR baby and managed accordingly.

Doppler abnormalities have been shown to deteriorate before biophysical profile scores (BPS) in the preterm fetus with IUGR prior to 32 weeks gestation.

6.2 Causes and risk factors for IUGR:

Review the medical and pregnancy history to determine the cause of the FGR e.g. accurate delivery date, normal anatomy scan, and if any history of infection.

Ensure all test results and 'management of care are clearly documented in HHR and DMR

Maternal	Fetal	Placental
Hypertensive disorders	Aneuploidy	Anatomical conditions
Autoimmune disease	Malformations	Vascular conditions
Certain medications	Abnormal genetic syndromes	Chromosomal conditions
Severe malnutrition, anaemia	Viral or protozoan infections	Morphological abnormalities
Maternal lifestyle e.g. smoking	Preterm birth	

alcohol abuse, substance abuse	Multiple gestation	
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6.3 Key points

1. An accurate expected delivery date (EDD) is a critical component to allow monitoring, assessment and optimal timing of delivery.
2. Management of the FGR fetus must include a balance of the risks of intrauterine chronic hypoxia with preterm delivery and its associated risks.
3. Fetal Doppler studies provide the most accurate non-invasive assessment for placental function. Absent or reversed UA Doppler's are associated with poor perinatal outcome and high perinatal mortality.

6.4 Diagnosis & definition:

Screening and diagnosis for FGR includes:

1. Accurate determination of the gestational age.
2. Abdominal palpation to determine fundal height during each antenatal visit.
3. Fundal Height (FH) measurements.
4. Ultrasound examination of a suspected SGA fetus.
5. Assessment of fetal well-being when an SGA fetus or FGR fetus is diagnosed. This includes Doppler studies, and cardiotocography monitoring (CTG) depending on gestation.
6. Any baby with EFW or AC $\leq 3^{\text{rd}}$ centile or baby with EFW or AC $\leq 10^{\text{th}}$ centile with abnormal dopplers is classified as FGR/IUGR baby or
7. Crossing centiles or a change in growth velocity AC/EFW crossing centile >2 quartiles is suggestive of FGR.

6.5 Management:

6.5.1 Determination of Gestational Age

A dating ultrasound in the first trimester provides the most accurate method to determine gestational age. If the earliest ultrasound was between 13 and 24 weeks of pregnancy and the last menstrual period (LMP) is certain, with regular menstruation, and there is a difference of less than 10 days between LMP & ultrasound, use the LMP estimate. If the LMP is uncertain or irregular menstruation, use the ultrasound EDD.

6.5.2 Abdominal Palpation

- The ability to detect fetal weight by palpation is limited. If there is suspicion of SGA, or FGR, management should be discussed with the obstetric team. A follow up ultrasound examination may be required.
- Document a management plan on the AKMR 70.5 'Obstetric Risk Assessment form after consultation with the GP Obstetrician or Consultant Obstetrician if an SGA or FGR fetus is suspected from palpation.

6.5.3 Fundal Height Measurements

- If SGA or FGR is suspected by abnormal Fundal Height (FH) measurements, ultrasound examination may be required after GP Obstetrician or Consultant Obstetrician consultation.

6.6 Fetal surveillance:

6.6.1 Ultrasound:

If there is suspicion of FGR ultrasound examination should be performed to assess:

- Anatomy examination - if an anatomy scan has not been done or is unavailable, this scan is required to exclude fetal anomalies, and fetal aneuploidy.
- Biometry – assessment of growth requires at least 2 measurements two weeks apart. Three weeks apart reduces false positive rate. Organise fortnightly biometry.
- Maximum vertical pocket (MVP). If MVP is increased, then to perform Amniotic Fluid Index (AFI). Amniotic fluid studies should be performed weekly along with Doppler studies.
- Doppler studies – Doppler studies are a valuable tool to differentiate between SGA fetus that is healthy and true IUGR fetus. Doppler studies to be performed weekly. If
 - o End Diastolic Velocities (EDV) present/ Pulsatility Index (PI) or Resistance Index (RI) >2SD: Arrange *twice-weekly* assessment of AFI and Doppler studies or more frequent surveillance if the clinically indicated.
 - o Absent / Reversed End Diastolic Velocities (A/REDV): Repeat UA and Ductus Venosus (DV) Doppler *daily*. Discuss with Obstetric Consultant or refer to Maternal Fetal Medicine at KEMH for further review depending on gestation age.

6.6.2 CTG Monitoring:

If the gestation is more than 32 weeks:

- Arrange twice weekly CTG in AAU
- If abnormal Doppler with A/REDV attend daily CTG while waiting for review at Tertiary Hospital. There should be a direct discussion between senior registrar or consultant for transfer of such patients.

If the gestation is less than 32 weeks gestation discuss with the Consultant on call if CTG monitoring is required

6.7 Further Management: (Appendix 1 and 3)

1. Frequency of fetal surveillance is assessed at each visit, and management plan adjusted by Obstetric team according to fetal and maternal clinical condition.
2. Antenatal surveillance may be conducted with antenatal clinic visits and by outpatient review in AAU. If the maternal or fetal clinical condition requires more intensive surveillance in-patient hospitalisation should be considered in consultation with the on-call Obstetrician.
3. All ultrasound examinations, CTGs, and BPP must be reviewed, and plan of management documented by the GP Obstetrician or Consultant Obstetrician.

Document the assessment and test results at each visit to AAU on the Maternity Attendance form AKMR8 and in the Digital Medical records (DMR).

6.8 Anticipated Preterm Birth

- Consider a course of corticosteroids if pre-term birth ≤ 34 weeks gestation is anticipated and birth is imminent within a week's time.
- Keep Paediatric team informed about any anticipated preterm delivery on the day. Give a brief history about the risk factors and the need for delivery.

6.9 Timing of Delivery

Delivery is indicated when risk of fetal death or morbidity is greater than the risk of prematurity.

6.9.1 Delivery with FGR/IUGR baby is indicated after 37 weeks if fetal surveillance is adequate and no further ultrasound deterioration in biometry and dopplers.

6.9.2 FGR with end diastolic flow present:

- If other surveillance findings and maternal condition are normal delivery may be delayed until 37 weeks.
- Recommend birth >34weeks if:
 - Static growth over 3-4 weeks
 - MCA Doppler PI <5th centile
- Consider steroids if caesarean birth.

6.9.3 IUGR associated with absent or reversed flow:

- Admit for close surveillance.
- Administration of steroids is recommended if preterm birth expected ≤ 34 weeks, if the clinical condition allows time.
- If other surveillance results are **abnormal** delivery is indicated.

6.10 Intrapartum management:

- Early admission in spontaneous labour.
- Apply continuous CTG monitoring from onset of uterine contractions.
- Caesarean birth is recommended in the IUGR fetus with UA A/REDV.
- Induction of labour can be offered where normal UA Doppler or abnormal UA PI with EDV present.

7. Legislative context

The following documents are required to give effect to this policy:

- Health Department of Western Australia (HDWA) [Recognising and Responding to Acute Deterioration Policy MP 0171/22](#)
- HDWA [Clinical Handover Policy MP 0095](#)
- HDWA [Cardiotocography Monitoring Policy MP 0076/18](#)
- East Metropolitan Health Service (EMHS) [Fetal Surveillance and Cardiotocography \(CTG\) Monitoring Policy EMHS: 195](#)

8. Supporting information

The following documents support the implementation of this policy:

- HDWA [Cardiotocography Monitoring Standard](#)
- AKG [Clinical Handover Procedure](#)
- AKG [Recognition and Response to Acute Deterioration Policy](#)
- AKG [Induction of Labour Guideline](#)

9. Compliance, monitoring and evaluation

Compliance with this guideline will be monitored via:

- Monitoring of reporting clinical incident data via Datix CIMS.
- Specific findings are to be reported at the relevant departmental meeting for review and identification of any opportunities for improvement

10. Relevant standards

National Safety and Quality Health Service (NSQHS) Standards (second edition):

- Standard 5 – Action 5.10 Screening of risk
- Standard 5 – Action 5.11 Clinical assessment
- Standard 6 – Action 6.7 Clinical handover
- Standard 6 – Action 6.9 Communicating critical information
- Standard 8 – Action 8.4 Recognising acute deterioration

11. References

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8. Marsal K. Obstetric management of intrauterine growth restriction. Best Practice & Research Clinical Obstetrics and Gynaecology. 2009;23:857-70.

12. Document owner

Enquiries relating to this guideline may be directed to:

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13. Review

This guideline will be reviewed and evaluated as required to ensure relevance and currency. At a minimum it will be reviewed within one (1) year after first issue and at least every three (3) years thereafter.

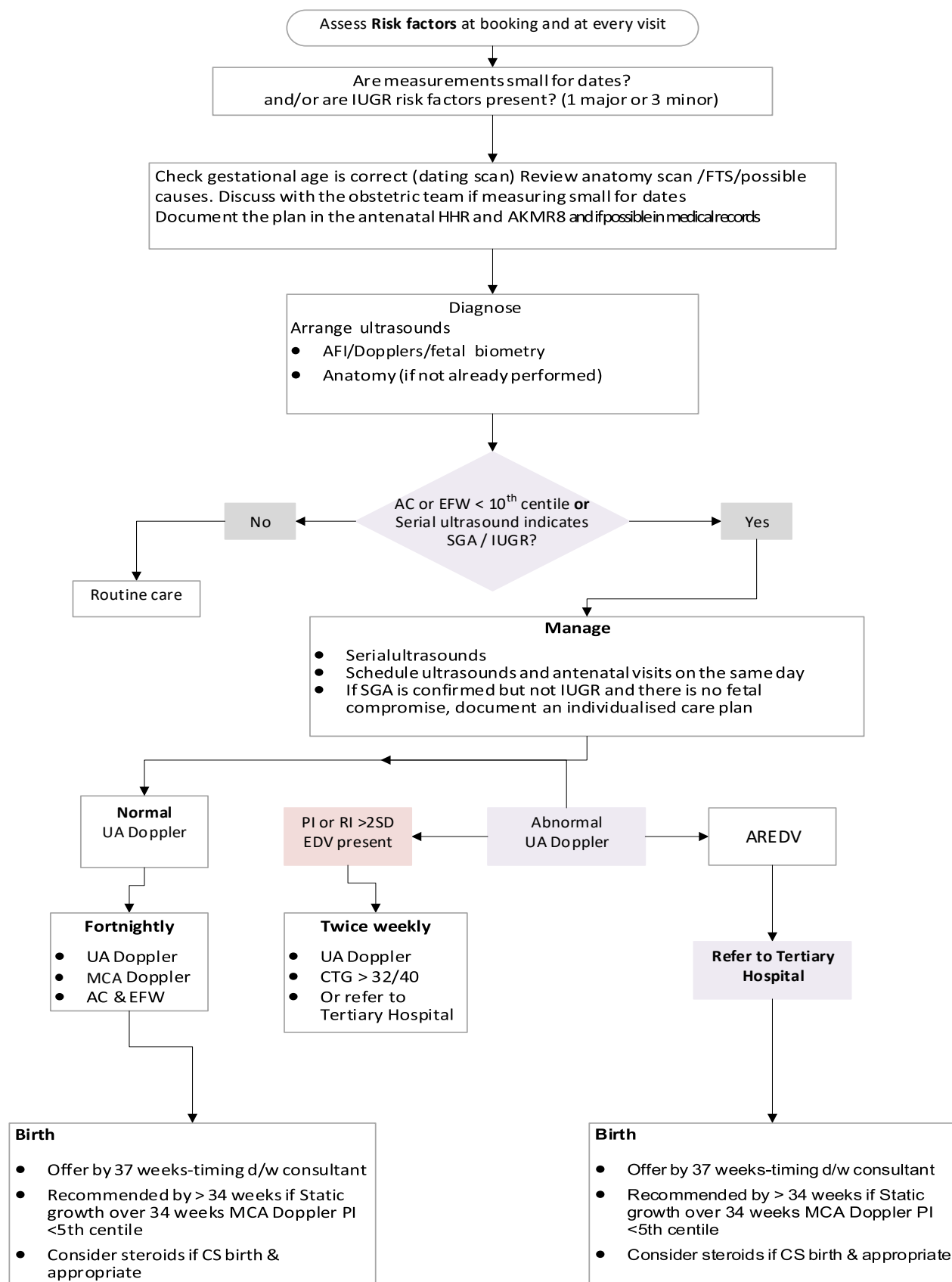
Version	Effective from	Effective to	Amendment(s)
V3	23/04/2024	23/04/2027	Minor amendment

14. Authorisation

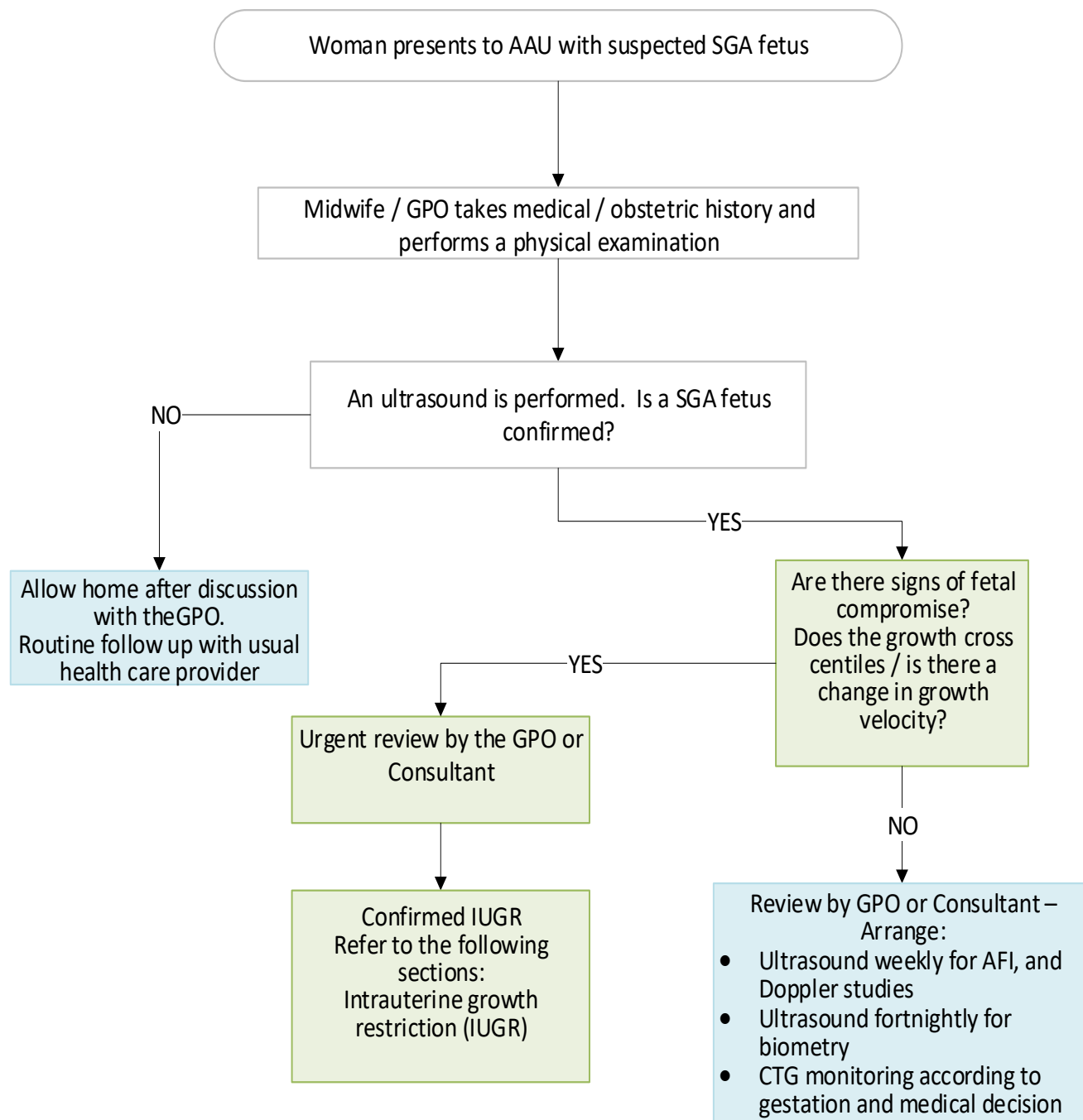
This guideline has been authorised and issued by:

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Appendix 1: Antenatal Assessment Unit Flowchart for Diagnosis and Management of IUGR



Appendix 2: Suspected Small for Gestational Fetus: Flow Chart



Appendix 3: Management of Confirmed FGR (IUGR) Flow Chart

